

AI Forex Trading System

EURUSD M5 Scalper

Daily User Guide

Session: 11:00 - 18:00 Manila | 03:00 - 10:00 UTC

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1 Quick Start

Run the system with a single command before your trading session:

```
cd ~/Personal/Trade  
python3 trade.py live
```

That's it. The live monitor opens a dashboard that refreshes every 60 seconds, fetching real EURUSD data from Yahoo Finance.

If you prefer an interactive menu:

```
python3 trade.py
```

All Commands

<code>python3 trade.py</code>	Interactive menu
<code>python3 trade.py live</code>	Start live monitor (main use)
<code>python3 trade.py status</code>	System health check
<code>python3 trade.py daily</code>	Full daily routine
<code>python3 trade.py backtest</code>	Run backtest
<code>python3 trade.py train</code>	Retrain ML model
<code>python3 trade.py download</code>	Refresh EURUSD data

2 Daily Workflow

Morning Setup (10:55 AM Manila)

Open a terminal and run the live monitor. It will wait until 11:00 AM

to start scanning for signals.

```
python3 trade.py live
```

During the Session (11:00 AM - 6:00 PM)

The dashboard updates every 60 seconds. Watch for signal arrows:

Signal	Meaning
^ BUY	Entry signal fired - price predicted to rise
v SELL	Entry signal fired - price predicted to fall
- SKIP	No signal - continue watching

End of Day (6:00 PM)

The session auto-closes. Press Ctrl+C to stop the monitor.

Optionally run the end-of-day check:

```
python3 trade.py daily
```

3 Understanding the Dashboard

```

=====
EURUSD Scalper | 14:25 Manila | Session: ACTIVE
=====
Signal:  ^ BUY
Score:   +0.623  (PA=+0.75, ML=0.710, conf=62%)
Entry:   1.08455
SL/TP:   1.08395 / 1.08755
Lot:     0.025

Positions: 1 open
          BUY 0.025L @ 1.08455  SL:1.08395 TP:1.08755

Balance:  $103.47
Today:    +$2.14 (+2.1%) | Total: +$3.47 (+3.5%)
Trades:   1/8 today | Refresh: 60s
=====

```

Dashboard Fields

- Signal: Entry direction (BUY / SELL / SKIP)
- Score: Combined signal strength (PA + ML + Sentiment)
- PA: Price action score from -1 (bearish) to +1 (bullish)
- ML: ML model confidence from 0.0 to 1.0. Gate requires >0.65
- Entry: Suggested entry price
- SL/TP: Stop loss and take profit levels
- Lot: Position size in lots (based on risk)
- Balance: Simulated account balance
- Today: Today's P&L and percentage
- Trades: Trade count today vs daily limit (8)

4 How Signals Work

The system uses a three-component fusion to generate signals:

1. Price Action (70% weight)

Scans M5 candles for RSI extremes, Bollinger Band touches, EMA pullbacks in trend, and range breakouts. This is the primary signal driver at 70% weight.

2. ML Model (15% weight)

XGBoost classifier trained on 16,927 M5 candles. Predicts if price will move 8+ pips in the next 10 minutes. Current accuracy: 57.8%. A confidence gate ($ML > 0.65$ or < 0.35) blocks low-confidence signals to improve win rate.

3. Sentiment (15% weight)

News headline sentiment via FinBERT. Currently 0 in simulator. Can be activated with a news feed.

Signal Flow

```

M5 Candle --> PA Score -->+
ML Model ---> ML Prob -->+---> Fuse --> Signal
Sentiment --> Score ----->+          (BUY/SELL/SKIP)
                                     |
                                     Confidence Gate
                                     (requires  $ML > 0.65$ )
                                     |
                                     Trade Execution
                                     (with SL/TP 1:4)
  
```

5 Risk Management

All risk controls are built-in and automatic:

Fixed Rules

- Max 8 trades per day
- Max 3% daily loss cap (auto-stops trading)
- Risk per trade: 1.5% of balance (scales with confidence up to 3%)
- Stop loss: 1.0 x ATR (typically 8-12 pips)
- Take profit: 4.0 x ATR (1:4 reward-to-risk)
- Session-restricted: 03:00 - 10:00 UTC only
- EURUSD only - single pair focus

Why 1:4 RR?

With a 37.7% win rate and 4:1 RR, the expected value per trade is:

$EV = (0.377 \times 4) - (0.623 \times 1) = 1.508 - 0.623 = +0.885 \text{ units}$
 This means every dollar risked returns \$0.885 on average.

Even at 25% win rate, the system breaks even with 4:1 RR.

Position Sizing

Lot size = Risk_USD / (SL_pips x \$10_per_pip_per_lot)

Example: \$100 balance, 1.5% risk (\$1.50), 10 pip SL

$Lot = \$1.50 / (10 \times \$10) = 0.015 \text{ lots} \Rightarrow \text{rounds to } 0.01 \text{ min}$

6 Performance Summary

Backtest: \$100 -> May 6 -> Jun 5 (22 trading days)

Win Rate:	37.7%
Profit Factor:	2.87
Final Balance:	\$280.65 (+180.7%)
Total Trades:	53
Max Drawdown:	5.9% (\$16.54)
Sharpe Ratio:	15.22

Walk-Forward (Out-of-Sample)

7 independent 5-day windows, each trained on prior 30 days only.

Zero look-ahead bias. All 7 windows were profitable.

Win Rate:	41.2%
Final Balance:	\$219.08 (+119%)
Profitable Windows:	7/7 (100%)

ML Model Accuracy

Original (14d training): 34.5%

Improved (60d + grid search): 57.8%

Features: 28 indicators (17 base + 11 added)

Best params: lookahead=2, threshold=8 pips,

1r

7 Files & Maintenance

Project Structure

trade.py	CLI launcher (main entry point)
main.py	Live monitor (yfinance data)
config.py	All settings in one place
train_models.py	ML model training
run_backtest.py	Backtest runner
walk_forward.py	Walk-forward validation
grid_search.py	Hyperparameter tuning
data/raw/	EURUSD M5 CSV data files
models/	Trained XGBoost model
logs/	Log files and trade records

Regular Maintenance

- Weekly: python3 trade.py download (refresh data)
- Every 2 weeks: python3 trade.py train (retrain model)
- After retraining: python3 trade.py backtest (verify)
- Check logs/system.log if something seems wrong

Configuration (config.py)

Key parameters you can tune:

BUY_THRESHOLD / SELL_THRESHOLD	Signal entry threshold
ML_CONFIDENCE_GATE	ML confidence gate
BASE_RISK_PCT	Risk per trade
ATR_MULT	SL/TP multipliers
MAX_DAILY_TRADES	Daily trade limit
MAX_DAILY_LOSS_PCT	Daily loss cap
TRADE_SESSION_START/END	Trading hours UTC

Important Notes

- System uses SIMULATOR mode (no real money)
- Yahoo Finance data has 0 volume for forex (clipped to 1)
- Session: 03:00-10:00 UTC = 11:00-18:00 Manila time
- MT5 not available on macOS - simulator only
- Best for: EURUSD scalping with 1:4 RR during active hours